The documentation and process conversion measures necessary to comply with this revision shall be completed by 21 February 2007.

INCH-POUND

MIL-PRF-19500/431D
21 November 2006
SUPERSEDING
MIL-PRF-19500/431C
11 October 2001

* PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTORS, N-CHANNEL, SILICON, TYPES 2N4091, 2N4092, 2N4093, 2N4091UB, 2N4092UB, AND 2N4093UB, JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for N-channel, junction, silicon field-effect transistors intended for use in chopper and analog gate circuit applications. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (similar to TO-18) and figure 2 (UB devices).

1.3 Maximum ratings. T_A = +25°C, unless otherwise specified.

Type	P_T (1) T_A = +25°C (free air)	V_{DS}	V_{DG}	V_{GS}	I_G	T_J	T_{STG}
	W	V dc	V dc	V dc	mA d	°C	°C
2N4091, UB	0.36	40	40	-40	10	-65 to +175	-65 to +200
2N4092, UB	0.36	40	40	-40	10	-65 to +175	-65 to +200
2N4093, UB	0.36	40	40	-40	10	-65 to +175	-65 to +200

(1) Derate linearly 2.4 mW/°C for T_A > +25°C.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.

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* 1.4 Primary electrical characteristics. T_C = +25°C, unless otherwise specified.

Type	$r_{ds(on)}$ V_{GS} = 0 I_D = 1 mA		Max I_{DSS} (1) V_{GS} = 0 V V_{DS} = 20 V dc		V_{DS}(on) maximum		
					I_D = 6.6 mA V_{GS} = 0	I_D = 4.0 mA V_{GS} = 0	I_D = 2.5 mA V_{GS} = 0
	Ω		mA dc		V dc	V dc	V dc
	Min	Max	Min	Max			
2N4091, UB		30	30		.20		
2N4092, UB		50	15			.20	
2N4093, UB		80	8				.20

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

* DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

* DEPARTMENT OF DEFENSE STANDARDS

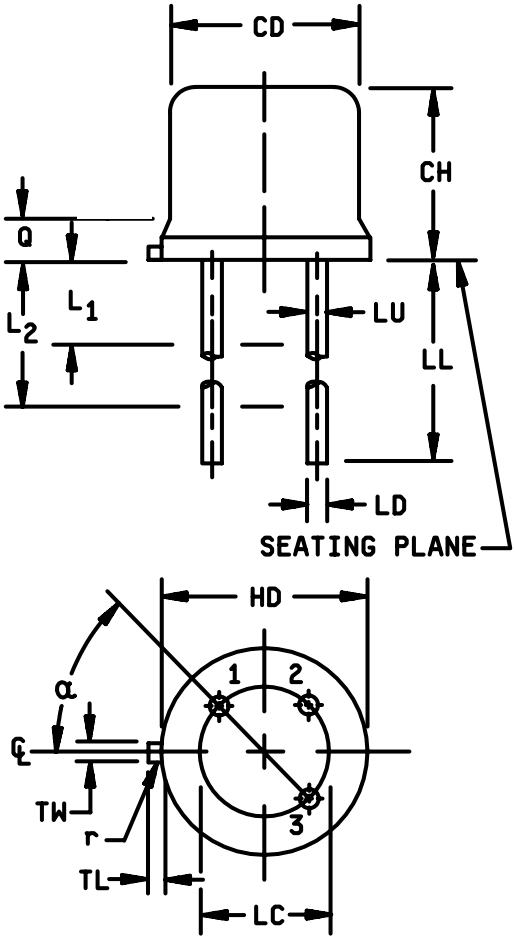
MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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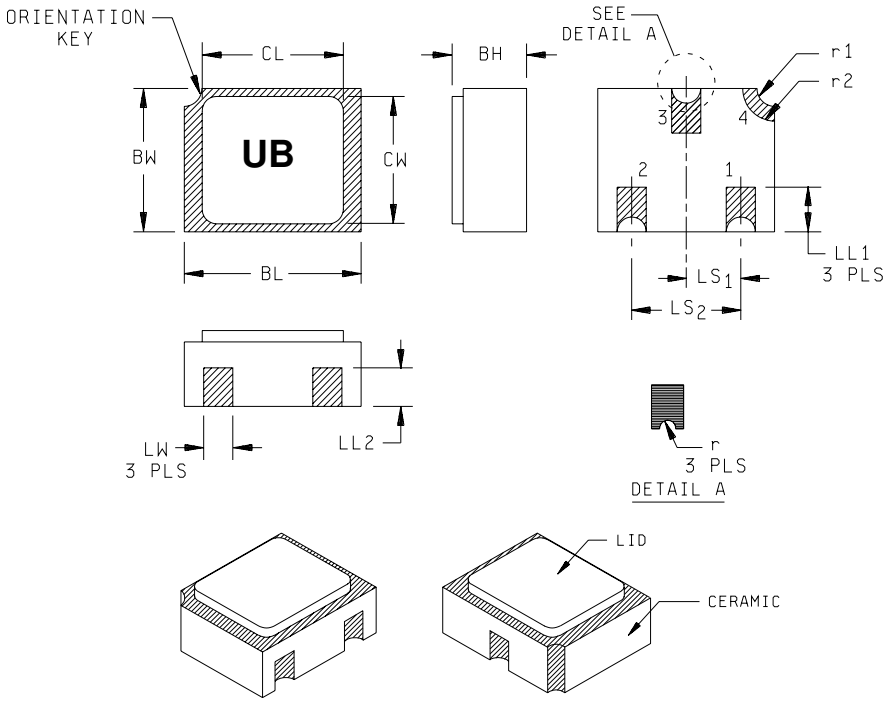
Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.170	.210	4.32	5.33	
HD	.209	.230	5.31	5.84	
LC	.100 TP		2.54 TP		6
LD	.016	.021	0.41	0.53	7,8
LL	.500	.750	12.7	19.05	7,8
LU	.016	.019	0.41	0.48	7,8
L_1		.050		1.27	7,8
L_2	.250		6.35		7,8
Q		.030		0.76	5
TL	.028	.048	0.71	1.22	3,4
TW	.036	.046	0.91	1.17	3
r		.010		0.25	10
α	45° TP		45° TP		6



NOTES:
1. Dimension are in inches.
2. Millimeters are given for general information only.
3. Beyond r (radius) maximum, TL shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.
12. Lead 1 = source, lead 2 = drain, lead 3 = gate.

FIGURE 1. Physical dimensions (similar to TO-18).

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BH	.046	.056	1.17	1.42
BL	.115	.128	2.92	3.25
BW	.085	.108	2.16	2.74
CL		.128		3.25
CW		.108		2.74
LL1	.022	.038	0.56	0.96
LL2	.017	.035	0.43	0.89
LS_1	.036	.040	0.91	1.02
LS_2	.071	.079	1.81	2.01
LW	.016	.024	0.41	0.61
r		.008		.203
r1		.012		.305
r2		.022		.559

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Hatched areas on package denote metallized areas.
4. Lid material: Kovar.
5. Pad 1 = Drain, Pad 2 = Source, Pad 3 = Gate, Pad 4 = Shielding connected to the lid.
6. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 2. Physical dimensions, surface mount (UB version).

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3. REQUIREMENTS

* 3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified MIL-PRF-19500, and figures 1 (similar to TO-18) and figure 2 (UB devices).

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction is not permitted to meet the requirements of this specification.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic discharge protection.

3.5.1 Handling. Metal oxide semiconductor (MOS) devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended (see 3.5).

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq$ or 100 kΩ, whenever bias voltage is applied drain to source.

* 3.6 Marking. Marking shall be in accordance with MIL-PRF-19500, except for the UB suffix package. Marking on the UB package shall consist of an abbreviated part number, the date code, and the manufacturer's symbol or logo. The prefixes JAN, JANTX, and JANTXV can be abbreviated as J, JX, and JV respectively. The "2N" prefix and the "UB" suffix can also be omitted.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.8 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and table I).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement
	JANTX and JANTXV levels
7	Optional.
9 , 10	Not applicable
11	I_{GSS1} , I_{DSS} , $r_{DS(on)}$
12	See 4.3.1.
13	Subgroup 2 of table I herein; ΔI_{GSS1} = ±50 pA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)}$ = ±20 percent of initial value. ΔI_{DSS} = ±20 percent of initial value.
14	Required.

* 4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: Method 1039 of MIL-STD-750, condition A; T_A = +175°C; V_{GS} = -24 V dc; V_{DS} = 0.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and 4.4.2.1 herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.2.1 Group B inspection table VIb, JAN, JANTX, and JANTXV of MIL-PRF-19500).

	Subgroup	Method	Condition
	B2	1051	Test condition G.
	B3	1027	Condition A, T_A = +175°C; V_{DS} = 0; V_{GS} = -24 V dc, 340 hrs.
*	B6	1032	T_{STG} = +200°C.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

	Subgroup	Method	Condition
*	C2	2036	Test condition E, not applicable to UB suffix devices.
	C6	1026	V_{GS} = -24 V dc; V_{DS} = 0; T_J = +175°C ±3°C.

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage, gate to source	3401	Bias condition C; V_{DS} = 0 V dc; I_G = -1.0 μA dc	$V_{(BR)GSS}$	-40		V dc
Gate reverse current	3411	V_{GS} = -20 V dc; V_{DS} = 0; bias condition C	I_{GSS1}		-0.1	nA dc
Drain current	3413	V_{DS} = 20 V dc; bias condition C	$I_{D(off)1}$		-0.1	nA dc
2N4091, UB		V_{GS} = -12 V dc				
2N4092, UB		V_{GS} = -8 V dc				
2N4093, UB		V_{GS} = -6 V dc				
Drain current	3413	V_{DS} = 20 V dc; V_{GS} = 0 (pulsed, see 4.5.1); bias condition C	I_{DSS}			
2N4091, UB				30		mA dc
2N4092, UB				15		mA dc
2N4093, UB				8		mA dc
Static drain to source on-state resistance	3421	V_{GS} = 0; pulsed (see 4.5.1); bias condition B; I_D = 1.0 mA dc	$r_{DS(on)}$			
2N4091, UB					30	Ω
2N4092, UB					50	Ω
2N4093, UB					80	Ω
Drain to source "on" state voltage	3405	V_{GS} = 0; bias condition B	$V_{DS(on)}$			
2N4091, UB		I_D = 6.6 mA dc			0.2	V dc
2N4092, UB		I_D = 4.0 mA dc			0.2	V dc
2N4093, UB		I_D = 2.5 mA dc			0.2	V dc

See footnote at end of table.

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TABLE I. Group A inspection - continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		T_A = +150°C				
Gate current	3411	Bias condition C; V_{DS} = 0 V dc V_{GS} = -20 V dc;	I_{GSS2}		-0.2	µA dc
Drain current	3413	V_{DS} = 20 V dc; bias condition A	$I_{D(off)2}$			
2N4091, UB		V_{GS} = -12 V dc			0.2	µA dc
2N4092, UB		V_{GS} = -8 V dc			0.2	µA dc
2N4093, UB		V_{GS} = -6 V dc			0.2	µA dc
Subgroup 4						
Small-signal common-source short-circuit input capacitance	3431	V_{DS} = 20 V dc; V_{GS} = 0; f = 1 MHz	C_{iss}		16	pF
Small-signal common-source reverse transfer capacitance	3433	V_{DS} = 0 V dc; V_{GS} = 20; f = 1 MHz	C_{rss}		5	pF
Turn-on delay time		See figure 3	$t_{d(on)}$			
2N4091, UB					15	ns
2N4092, UB					15	ns
2N4093, UB					15	ns
Rise time		See figure 3	t_r			
2N4091, UB					10	ns
2N4092, UB					20	ns
2N4093, UB					40	ns
Turn-off delay time		See figure 3	$t_{d(off)}$			
2N4091, UB					40	ns
2N4092, UB					60	ns
2N4093, UB					80	ns
Subgroups 5, 6, 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

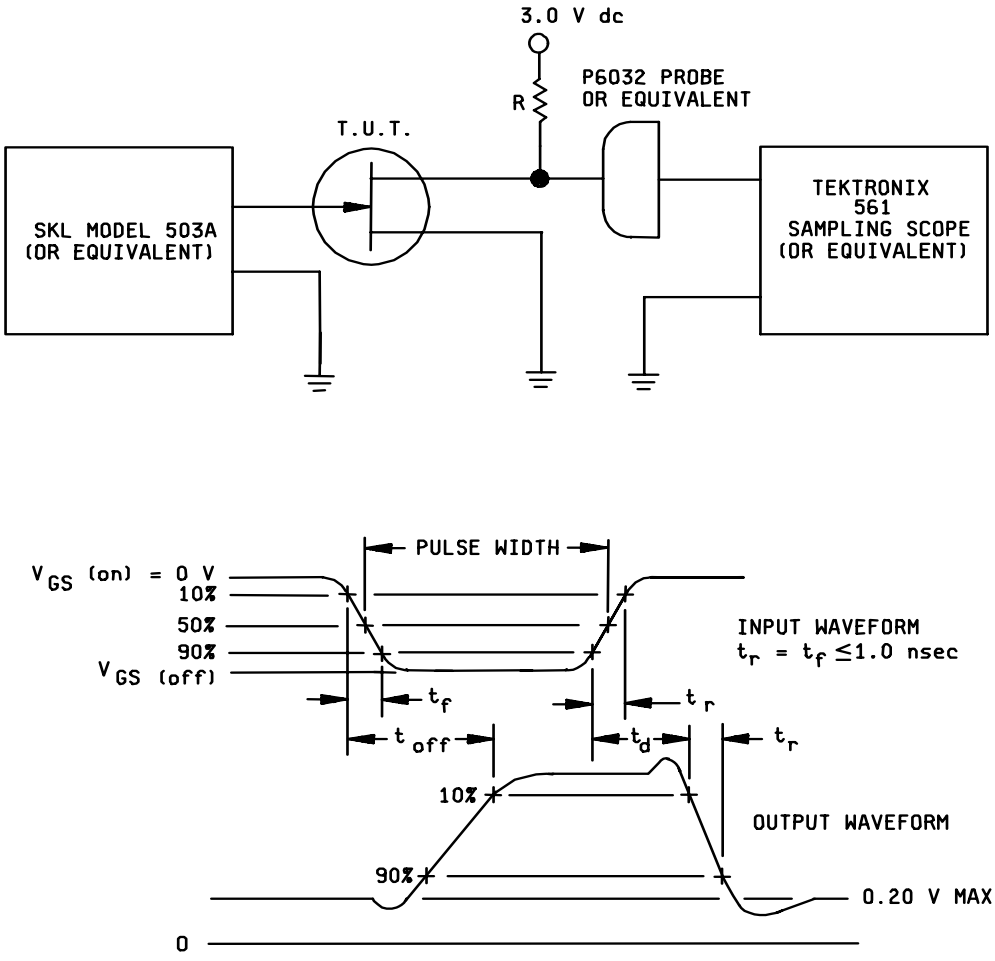
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* TABLE II. Group E inspection (all quality levels) for qualification or re-qualification only.

Inspection	MIL-STD-750		Sample size
	Method	Conditions	
Subgroup 1 1/			45 devices c = 0
Temperature cycling	1051	-55°C to +150°C, 500 cycles	
Hermetic seal Fine leak Gross leak	1071	 Test conditions G or H Test conditions C or D	
Electrical measurements		See table I, subgroup 2	
Subgroup 2 1/			45 devices c = 0
Blocking life	1048	1,000 hours minimum, T_A = +150°C, V_{DG} or V_{GS} = 80 percent of rated.	
Electrical measurements		See table I, subgroup 2	
Subgroup 4			
Not applicable			
Subgroup 6			3 devices
ESD	1020		

1/ A separate sample for each test shall be pulled.

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NOTES:

1. Measure under pulse conditions: Pulse width ≤ 300 ms; duty cycle ± 3 percent.
2. 2N4091, 2N4091UB: I_D = 6.6 mA.
   2N4092, 2N4092UB: I_D = 4.0 mA.
   2N4093, 2N4093UB: I_D = 2.5 mA.
3. 2N4091, 2N4091UB: $V_{GS(off)}$ = -12 V.
   2N4092, 2N4092UB: $V_{GS(off)}$ = -8.0 V.
   2N4093, 2N4093UB: $V_{GS(off)}$ = -6.0 V.

* FIGURE 3. Switching time test circuit and waveforms.

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

* 6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Air Force - 19, 71, 99

Preparing activity:
DLA - CC

(Project 5961-2006-007)

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